

ABHISHEK KAR

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SUMMARY

Mechanical Engineer (MSME, 4+ years) owning full product design from concept through NPI, validation, and production ramp for semiconductor capital equipment and instrumentation built to consumer-electronics-class tolerances. Deep in SolidWorks, Siemens NX, and CATIA, with GD&T, statistical tolerance analysis, FEA, and hands-on prototyping and fabrication.

- **Full Product Design Ownership:** Own designs from concept to production across 8+ programs, cutting rework 31%.
- **Precision & Opto-Mechanical Design:** Design opto-mechanical, electromechanical, and enclosure hardware for RF/plasma.
- **Statistical Tolerance & Root Cause:** Traced an alignment failure to the datum scheme, cutting build issues 25%.
- **Hands-On Prototyping & DFM:** Build by hand via CNC, welding, and 3D printing; design for molding and sheet metal.

SKILLS

Design and CAD: SolidWorks, Siemens NX, Creo, CATIA, Autodesk Inventor (Certified), opto-mechanical design, precision assemblies, enclosure design, GD&T (ASME Y14.5), 2D drawings

Analysis and Simulation: FEA (structural, thermal, modal), ANSYS, CFD (ANSYS Fluent), Monte Carlo simulation, tolerance stack-up, statistical tolerance analysis, dimensional variation analysis, first-principles modeling, MATLAB

Manufacturing and Prototyping: DFM, DFA, hands-on prototyping and fabrication, CNC machining, lathe, mill, welding, 3D printing, injection molding, sheet metal, NPI, concept to production

Quality and Collaboration: DFMEA, PFMEA, root cause and corrective action, SPC (Cp/Cpk), DOE, reliability, validation testing, supplier coordination/FAI, BOM and ECO, SAP-PLM, Teamcenter, Jira, Agile, OTD, MS Office (MS Word, MS Excel, MS Project), cross-functional collaboration

WORK EXPERIENCE

L&T Technology Services (Client: Advanced Energy)

Jun 2025 - Present

Senior Mechanical Engineer

New York, USA

- Designed **opto-mechanical** and **electromechanical** assemblies and fixtures for RF and plasma semiconductor equipment in **SolidWorks** and **Siemens NX** from concept to production across 8+ programs, cutting rework 31%.
- Performed **Monte Carlo tolerance simulation** and **dimensional variation analysis** on 10+ precision assemblies, redesigning a datum scheme through **root cause** and cutting alignment-related build issues 25%.
- Generated detailed **2D drawings** with **GD&T** for production and inspection and processed 15+ **ECOs** in **SAP-PLM**, improving build accuracy 20%.
- Partnered with **suppliers, manufacturing, and quality** to ensure producible, reliable designs, cutting supplier lead time 23%.
- Led **BOM validation** and debugging across 30+ builds and supported **design validation**, defining **test criteria** and troubleshooting hardware at the assembly and system levels.
- Saved ~\$120K by streamlining design task tracking and **cross-functional** issue resolution through **Jira-based Agile workflows**, improving escalation, reporting, and **on-time delivery (OTD)** to commit.

University of Minnesota

Jul 2024 - Jun 2025

Mechanical Engineer

Minnesota, USA

- Developed **structural, thermal, and modal FEA** in **ANSYS** characterizing heat transfer, stress, and vibration, validating against analytical benchmarks.
- Executed **CFD** in **ANSYS Fluent** with **k-epsilon turbulence** closure, deriving governing equations from **first principles** to support thermal and fluid design decisions.

Emerson / University of Minnesota

Sep 2023 - May 2024

Product Development Engineer

Minnesota, USA

- Built and validated 11 pre-production units by hand through **machining, welding, and 3D printing**, iterating **CAD** from test data and cutting build time 17%.
- Applied **DFM** and **DFA** for machined and molded components with manufacturing and industrial design, reducing design revisions from 7 to 3 per program.
- Conducted **FEA** and **DFMEA** to optimize structural integrity and thermal performance, improving product reliability and durability across the flow-meter line.

BGR Mining and Infra Limited

Jun 2020 - Jul 2022

Mechanical Engineer

Odisha, India

- Led **root-cause failure analysis** and **corrective action** across 20+ production equipment systems, cutting incidents 15% and improving reliability and operability.
- Engineered **Lean** and **OEE** programs using **Minitab** to standardize processes, cutting downtime 6% and saving roughly \$50K per year.

EDUCATION

University of Minnesota

Sep 2022 - May 2024

Master of Science, Mechanical Engineering

Minnesota, USA

KIIT University

Aug 2013 - May 2017

Bachelor of Technology, Mechanical Engineering

Odisha, India